1 Theoretical Fundation

The exercise consists of determining the position of objects in space, depeding on the distance between them.

Considering a matrix:

$$\boldsymbol{D} = \begin{pmatrix} 0 & d_{12} & ... & d_{1n} \\ d_{21} & 0 & ... & d_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ d_{n1} & d_{n2} & ... & 0 \end{pmatrix} \tag{1}$$

which represents the pairwise distance between the objects. The Matrix is symmetric.

The objective is to determine the position of the objects, in this case 2 dimensional, represented with the matrix:

$$\boldsymbol{X}_{2D} = \begin{pmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \\ \vdots & \vdots \\ x_{n1} & x_{n2} \end{pmatrix}. \tag{2}$$

We can reformulate the position array as 1D data vector:

$$\boldsymbol{X} = \begin{pmatrix} x_{11} \\ x_{12} \\ x_{21} \\ \vdots \\ x_{n1} \\ x_{n2} \end{pmatrix}. \tag{3}$$

1.1 Optimization Problem

The objective is to find $\boldsymbol{X}_{2D}$ or $\boldsymbol{X}$, so that the the distance between the positions is approximatly the value of the $\boldsymbol{D}$ matrix.

Meaning, the objective is:

$$\min_{\boldsymbol{X}_{2D}} h_{\boldsymbol{X}_{2D}}(\boldsymbol{D}) = \sum_{i=2}^{n} \sum_{j=1}^{i-1} |d_{ij} - \|\boldsymbol{x}_{2D,i} - \boldsymbol{x}_{2D,j}\|| \tag{4}$$

Can be changed to use $\boldsymbol{X}$.

$$\min_{\boldsymbol{X}} h_{\boldsymbol{X}}(\boldsymbol{D}) = \sum_{i=2}^{n} \sum_{j=1}^{i-1} \left| d_{ij} - \sqrt{(x_{i1} - x_{j1})^2 + (x_{i2} - x_{j2})^2} \right| \tag{5}$$

Since the absolute of a value is not a differentiable value, the square is used.

$$\min_{\boldsymbol{X}} h_{\boldsymbol{X}}(\boldsymbol{D}) = \sum_{i=2}^{n} \sum_{j=1}^{i-1} \left(d_{ij} - \sqrt{(x_{i1} - x_{j1})^2 + (x_{i2} - x_{j2})^2} \right)^2 \tag{6}$$

The term is defined as:

$$f_{ij}(\boldsymbol{D}, \boldsymbol{X}) = \left(d_{ij} - \sqrt{(x_{i1} - x_{j1})^2 + (x_{i2} - x_{j2})^2} \right) \tag{7}$$

Differentiating $h_{\boldsymbol{X}}$ gives the term:

$$\frac{\partial}{\partial x_{kl}} h_{\boldsymbol{X}}(\boldsymbol{D}) = \sum_{j=1}^{k-1} \frac{\partial}{\partial x_{kl}} f_{kj}(\boldsymbol{D}, \boldsymbol{X}) + \sum_{i=k+1}^{n} \frac{\partial}{\partial x_{kl}} f_{ik}(\boldsymbol{D}, \boldsymbol{X}) \tag{8}$$

And differentiating f_{ij}:

$$\begin{aligned}\frac{\partial}{\partial x_{kl}} f_{ij}(\boldsymbol{D}, \boldsymbol{X}) = & \\ & 2\left(d_{ij} - \sqrt{(x_{i1} - x_{j1})^2 + (x_{i2} - x_{j2})^2}\right) \\ & \left(-\frac{1}{2\sqrt{(x_{i1} - x_{j1})^2 + (x_{i2} - x_{j2})^2}}\right) \\ & \begin{cases} k = i, l = 1 & (2(x_{i1} - x_{j1})) \\ k = i, l = 2 & (2(x_{i2} - x_{j2})) \\ k = j, l = 1 & (-2(x_{i1} - x_{j1})) \\ k = j, l = 2 & (-2(x_{i2} - x_{j2})) \\ \text{otherwise} & 0 \end{cases}\end{aligned} \tag{9}$$

Differentiation again $\frac{\partial}{\partial x_{kl}} h_{\boldsymbol{X}}(\boldsymbol{D})$ gives:

$$\begin{aligned}\frac{\partial^2}{\partial x_{kl} \partial x_{qr}} h_{\boldsymbol{X}}(\boldsymbol{D}) = & \\ & \begin{cases} q < k & \frac{\partial^2}{\partial x_{kl} \partial x_{qr}} f_{km}(\boldsymbol{D}, \boldsymbol{X}) \\ q = k & \sum_{j=1}^{k-1} \frac{\partial^2}{\partial x_{kl} \partial x_{qr}} f_{kj}(\boldsymbol{D}, \boldsymbol{X}) \\ q > k & 0 \end{cases} \\ & + \begin{cases} q < k & 0 \\ q = k & \sum_{i=k+1}^{n} \frac{\partial^2}{\partial x_{kl} \partial x_{qr}} f_{ik}(\boldsymbol{D}, \boldsymbol{X}) \\ q > k & \frac{\partial^2}{\partial x_{kl} \partial x_{qr}} f_{mk}(\boldsymbol{D}, \boldsymbol{X}) \end{cases}\end{aligned} \tag{10}$$

and $\frac{\partial^2}{\partial x_{kl}\partial x_{qr}} f_{ij}(\boldsymbol{D},\boldsymbol{X})$:

$$
\begin{aligned}
&\frac{\partial^2}{\partial x_{kl}\partial x_{qr}} f_{ij}(\boldsymbol{D},\boldsymbol{X}) = \\
&\qquad 2\left(-\frac{1}{2\sqrt{(x_{i1}-x_{j1})^2+(x_{i2}-x_{j2})^2}}\right) \\
&\qquad \begin{cases} q=i,r=1 & (2(x_{i1}-x_{j1})) \\ q=i,r=2 & (2(x_{i2}-x_{j2})) \\ q=j,r=1 & (-2(x_{i1}-x_{j1})) \\ q=j,r=2 & (-2(x_{i2}-x_{j2})) \\ \text{otherwise} & 0 \end{cases} \\
&\qquad \left(-\frac{1}{2\sqrt{(x_{i1}-x_{j1})^2+(x_{i2}-x_{j2})^2}}\right) \\
&\qquad \begin{cases} k=i,l=1 & (2(x_{i1}-x_{j1})) \\ k=i,l=2 & (2(x_{i2}-x_{j2})) \\ k=j,l=1 & (-2(x_{i1}-x_{j1})) \\ k=j,l=2 & (-2(x_{i2}-x_{j2})) \\ \text{otherwise} & 0 \end{cases} \\
&\qquad +2\left(d_{ij}-\sqrt{(x_{i1}-x_{j1})^2+(x_{i2}-x_{j2})^2}\right) \\
&\qquad \left(\frac{1}{4\left((x_{i1}-x_{j1})^2+(x_{i2}-x_{j2})^2\right)^{\frac{3}{2}}}\right) \\
&\qquad \begin{cases} q=i,r=1 & (2(x_{i1}-x_{j1})) \\ q=i,r=2 & (2(x_{i2}-x_{j2})) \\ q=j,r=1 & (-2(x_{i1}-x_{j1})) \\ q=j,r=2 & (-2(x_{i2}-x_{j2})) \\ \text{otherwise} & 0 \end{cases} \\
&\qquad \begin{cases} k=i,l=1 & (2(x_{i1}-x_{j1})) \\ k=i,l=2 & (2(x_{i2}-x_{j2})) \\ k=j,l=1 & (-2(x_{i1}-x_{j1})) \\ k=j,l=2 & (-2(x_{i2}-x_{j2})) \\ \text{otherwise} & 0 \end{cases} \\
&\qquad +2\left(d_{ij}-\sqrt{(x_{i1}-x_{j1})^2+(x_{i2}-x_{j2})^2}\right) \\
&\qquad \left(-\frac{1}{2\sqrt{(x_{i1}-x_{j1})^2+(x_{i2}-x_{j2})^2}}\right) \\
&\qquad \begin{cases} k=i,l=1,q=i,r=1 & (2) \\ k=i,l=1,q=j,r=1 & (-2) \\ k=i,l=2,q=i,r=2 & (2) \\ k=i,l=2,q=j,r=2 & (-2) \\ k=j,l=1,q=i,r=1 & (-2) \\ k=j,l=1,q=j,r=1 & (2) \\ k=j,l=2,q=i,r=2 & (-2) \\ k=j,l=2,q=j,r=2 & (2) \end{cases}
\end{aligned}
\tag{11}
$$

The term $\frac{\partial^2}{\partial x_{kl}\partial x_{qr}} f_{ij}(\boldsymbol{D},\boldsymbol{X})$ can be resumed to:

$$
\begin{aligned}
\frac{\partial^2}{\partial x_{kl}\partial x_{qr}} f_{ij}(\boldsymbol{D},\boldsymbol{X}) = & \\
& \left[\left(\frac{1}{2\left((x_{i1}-x_{j1})^2+(x_{i2}-x_{j2})^2\right)}\right)+\left(\frac{d_{ij}-\sqrt{(x_{i1}-x_{j1})^2+(x_{i2}-x_{j2})^2}}{2\left((x_{i1}-x_{j1})^2+(x_{i2}-x_{j2})^2\right)^{\frac{3}{2}}}\right)\right] \\
& \begin{cases} q=i, r=1 & (2(x_{i1}-x_{j1})) \\ q=i, r=2 & (2(x_{i2}-x_{j2})) \\ q=j, r=1 & (-2(x_{i1}-x_{j1})) \\ q=j, r=2 & (-2(x_{i2}-x_{j2})) \\ \text{otherwise} & 0 \end{cases} \\
& \begin{cases} k=i, l=1 & (2(x_{i1}-x_{j1})) \\ k=i, l=2 & (2(x_{i2}-x_{j2})) \\ k=j, l=1 & (-2(x_{i1}-x_{j1})) \\ k=j, l=2 & (-2(x_{i2}-x_{j2})) \\ \text{otherwise} & 0 \end{cases} \\
& + \left(-\frac{d_{ij}-\sqrt{(x_{i1}-x_{j1})^2+(x_{i2}-x_{j2})^2}}{\sqrt{(x_{i1}-x_{j1})^2+(x_{i2}-x_{j2})^2}}\right) \\
& \begin{cases} k=q, l=r & 2 \\ k\neq q, l=r & -2 \\ k,q \neq \{i,j\} & 0 \\ \text{otherwise} & 0 \end{cases}
\end{aligned} \tag{12}
$$

To ensure convexity, $\frac{\partial^2}{\partial x_{kl}\partial x_{qr}} h_{\boldsymbol{X}}(\boldsymbol{D}) \geq 0 \qquad \forall \quad k,q \in [1,n]; \quad l,r \in \{1,2\}$

- $q<k: \quad \frac{\partial^2}{\partial x_{kl}\partial x_{qr}} f_{kq}(\boldsymbol{D},\boldsymbol{X}) =$

$$
\left[\left(\frac{1}{2((x_{i1}-x_{j1})^2+(x_{i2}-x_{j2})^2)}\right)+\left(\frac{d_{ij}-\sqrt{(x_{i1}-x_{j1})^2+(x_{i2}-x_{j2})^2}}{2((x_{i1}-x_{j1})^2+(x_{i2}-x_{j2})^2)^{\frac{3}{2}}}\right)\right] \begin{cases} q=j, r=1 & (-2(x_{i1}-x_{j1})) \\ q=j, r=2 & (-2(x_{i2}-x_{j2})) \\ \text{otherwise} & 0 \end{cases} \begin{cases} k=i, l=1 & (2(x \\ k=i, l=2 & (2(x \\ \text{otherwise} & 0 \end{cases}
$$

$$
\left(\frac{d_{ij}-\sqrt{(x_{i1}-x_{j1})^2+(x_{i2}-x_{j2})^2}}{\sqrt{(x_{i1}-x_{j1})^2+(x_{i2}-x_{j2})^2}}\right) \begin{cases} l=r & -2 \\ \text{otherwise} & 0 \end{cases}
$$